

AI-Powered Zero-Day Attack Detection Platform — Evaluation Report

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1. Dataset Overview

Dataset: Sample IDS Dataset

Known classes (used for training): benign, probe, r2l, u2r, worm

Unknown / held-out classes (zero-day simulation): dos

2. Model Comparison

All models were trained on identical known-class splits and evaluated with the same held-out unknown classes to ensure a fair comparison.

Model	Acc.	Prec.	Recall	F1	MCC	ROC-AUC	FPR	Unk. Recall	Train (s)	Infer (ms)
cnn	0.896	0.886	0.815	0.842	0.774	0.971	0.032	0.948	1.5	2.72
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3. Key Finding

cnn achieved the highest unknown-attack recall (0.948), making it the strongest candidate for zero-day detection among the architectures compared.